

# Human Evaluation Protocol

August 3, 2026

We compare generated videos using human assessors and report the Mean Opinion Score (MOS). Each assessor rates videos on a fixed scale, for example from 1 to 5, where higher values indicate more realistic and task-consistent skill execution. We also conduct pairwise comparisons to select the best model. We ask experts to choose which model performs better or whether they are of equal quality. To do this, we present two generated videos alongside a real video. Videos are anonymized and randomly ordered so that assessors do not know which World Model (WM) condition generated each sample.

To evaluate the model’s performance across five manipulation tasks with 200 examples each, we employ two parallel approaches: subjective quality testing via the MOS method with 14 assessors, and absolute discrete rating with 25 assessors using a 3-person overlap to ensure data reliability. For more precise measurements, each video is evaluated along three dimensions: visual quality, gripper trajectory following, and compliance with physical constraints. The pairwise comparison uses the same three dimensions, but assessors select which generated video is closer to the real reference for each criterion.

## 1 MOS Rating Criteria

- **Visual quality assessment.**

1. Unusable: Critical degradation in quality (unable to determine the shape or color of objects, severe artifacts or dropped frames); key elements of the scene are not identifiable; video is unsuitable for analysis; severe color distortion possible; complete loss of visual information.
2. Poor: Significant visual defects (severe blur, over- or underexposure, image instability); artifacts or dropped frames interfere with

perception; important elements (e.g., gripper) may become indistinguishable; difficult to interpret state (e.g., gripper open/closed).

3. Fair: Noticeable issues (moderate blur, unstable autofocus/exposure, noise); possible temporary degradation in quality; manipulated object is partially blurred but remains discernible; scene as a whole is interpretable.
4. Good: Minor defects (slight blur, slight exposure variations, low noise); overall scene visibility is preserved.
5. Excellent: High clarity and stable images; correct exposure and color balance; no noticeable artifacts; all key elements are always visible; correct object color and shape preservation; no "floating" textures.

- **Gripper trajectory following quality assessment.**

1. Not Completed: Instructions not followed; actions inconsistent with or contradictory to the task; video static or uninformative.
2. Minimally Completed: A significant portion of the task not completed; movement poorly follows instructions; goals not achieved.
3. Partially Completed: Only the main components are completed; noticeable errors or omissions present; 75
4. Almost Completed: Most instructions are followed; minor deviations do not affect the result; 90
5. Fully Completed: Instructions are followed accurately and completely; all key actions are correct; the gripper always reaches the goal and opens/closes correctly.

- **Physical law compliance assessment.**

1. Physically impossible: Gross physics violations (teleportation, movement without contact, lack of cause-and-effect relationships); significant geometric inconsistencies; if a human performs the task instead of a robot, automatically rated 1.
2. Poorly physically based: Frequent or serious physics violations; incorrect interactions; penetrations through objects; unrealistic movement.
3. Moderately physically based: Noticeable but non-critical violations; movements look unnatural in places; partial dynamic deviations.

4. Mostly physically based: Minor deviations from real physics; overall behavior remains plausible; no critical errors.
5. Fully physically based: All movements are realistic and consistent; interactions comply with physics; no penetration artifacts; static videos (without movement) also receive a 5.

## 2 Pairwise Comparison Criteria

For pairwise comparison, assessors select the generated video (A or B) that is closest to the real reference for one criterion at a time.

- **Visual quality assessment.** Assessors select the video closest to the reference in visual quality while ignoring trajectory following and physicality of interaction. The criteria are: clarity and detail, visual artifacts, object consistency, and correct object behavior under occlusion.
- **Gripper trajectory following quality assessment.** Assessors select the video that most closely replicates the robotic arm trajectory from the reference while ignoring visual quality and physicality of interaction. The criteria are: movement direction, goal reaching, gripper control, execution completeness, and whether the gripper is closest to the object when both videos fail.
- **Physical law compliance assessment.** Assessors select the video that is most plausible in terms of real-world physics while ignoring visual quality and trajectory following. The criteria are: correct contact with objects and surfaces, geometric correctness, cause-and-effect relationships, and natural behavior after object release.

Critical physical errors include: object motion without contact, objects sticking to the gripper without being grasped, passing through solid surfaces, and missing reaction after release.